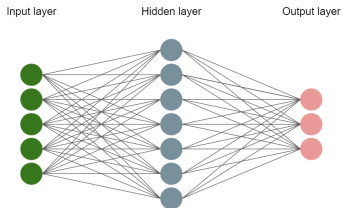
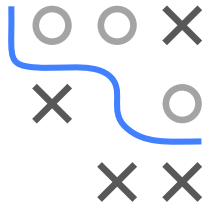


Introduction to Machine Learning

ML-Basics

Models & Parameters



Learning goals

- Understand that an ML model is simply a parametrized function
- Understand that the hypothesis space lists all admissible models
- Understand relationship between hypothesis and parameter space

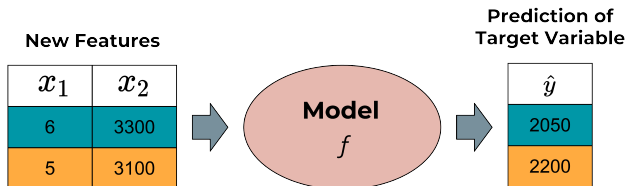
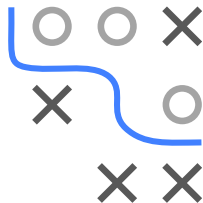
WHAT IS A MODEL?

- A **model** (or **hypothesis**)

$$f : \mathcal{X} \rightarrow \mathbb{R}^g$$

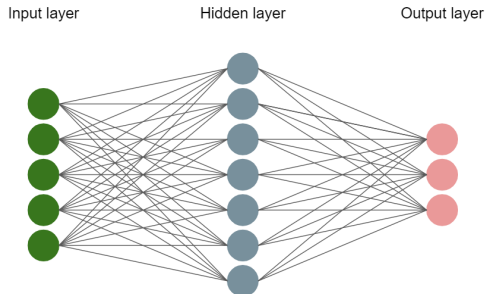
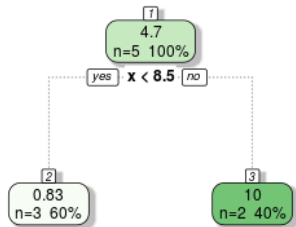
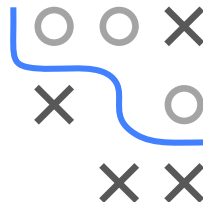
is a function that maps feature vectors to predicted target values

- In regression: $g = 1$; in classification, g is the number of classes, and output vectors are scores or class probabilities



WHAT IS A MODEL?

- f is meant to capture intrinsic patterns of the data, the underlying assumption being that these hold true for *all* data drawn from $\mathbb{P}_{xy}$
- Models can range from super simple (e.g., linear, tree stumps) to very complex (e.g., DL) with lots of choices

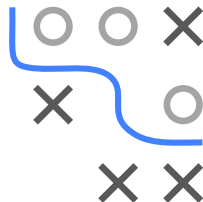


- ML requires **constraining** f to a certain type of functions

PARAMETRIZATION

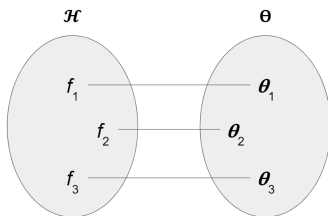
- All models within a hypothesis space share a common functional structure typically constructed as **parametrized family of functions**
- We collect all parameters in a **parameter vector** $\theta = (\theta_1, \theta_2, \dots, \theta_d)$ from **parameter space** Θ
- They are our means of fixing a specific function from the family: once set our model is fully determined
- Therefore, we can re-write $\mathcal{H}$ as:

$$\mathcal{H} = \{f_{\theta} : f_{\theta} \text{ belongs to a certain functional family} \\ \text{parameterized by } \theta\}$$

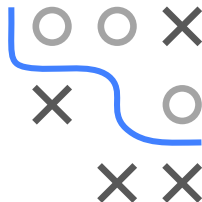


PARAMETRIZATION

- Finding optimal model = finding optimal parameters
- This allows us to operationalize our search for the best model as a search for the optimal value on a d -dimensional parameter surface

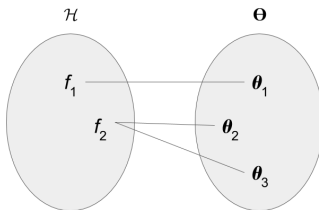
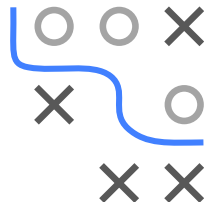


- θ might be scalar or very high-dimensional with thousands of parameters depending on the complexity of our model



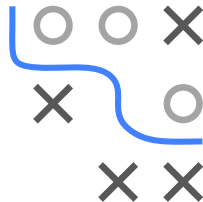
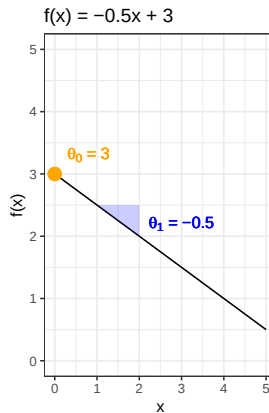
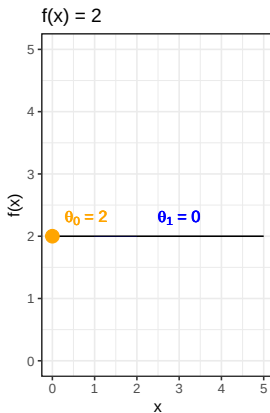
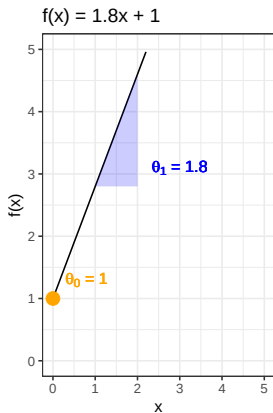
PARAMETRIZATION

- Some parameter vectors, for some model classes, encode the same function: the parameter-to-model mapping could be non-injective
- We call this a non-identifiable model
- This shall not concern us here

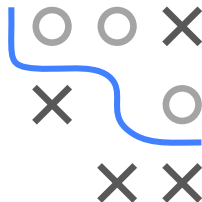


EXAMPLE: UNIVARIATE LINEAR FUNCTIONS

$$\mathcal{H} = \{f : f(\mathbf{x}) = \theta_0 + \theta_1 x, \theta \in \mathbb{R}^2\}$$

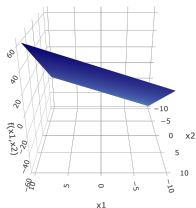


EXAMPLE: BIVARIATE QUADRATIC FUNCTIONS

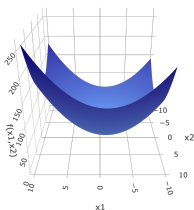


$$\mathcal{H} = \{f : f(\mathbf{x}) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2 + \theta_5 x_1 x_2, \theta \in \mathbb{R}^6\}$$

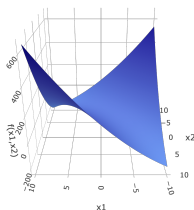
$$f(x) = 3 + 2x_1 + 4x_2$$



$$f(x) = 3 + 2x_1 + 4x_2 + 1x_1^2 + 1x_2^2$$



$$f(x) = 3 + 2x_1 + 4x_2 + 1x_1^2 + 1x_2^2 + 4x_1 x_2$$



EXAMPLE: RBF NETWORK

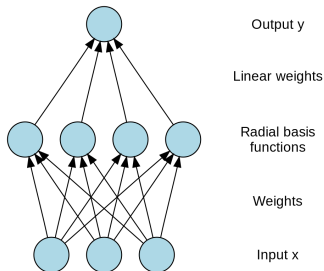
Radial basis function networks with Gaussian basis functions

$$\mathcal{H} = \left\{ f : f(\mathbf{x}) = \sum_{i=1}^k a_i \rho(\|\mathbf{x} - \mathbf{c}_i\|) \right\}$$

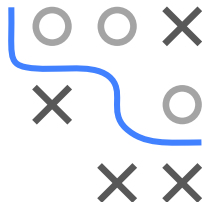
where

- a_i is the weight of the i -th neuron
- $\mathbf{c}_i$ its center vector and
- $\rho(\|\mathbf{x} - \mathbf{c}_i\|) = \exp(-\beta\|\mathbf{x} - \mathbf{c}_i\|^2)$ is the i -th radial basis function with bandwidth $\beta \in \mathbb{R}$

Usually number of centers k and bandwidth β need to be set in advance (so-called *hyperparameters*)



► [Click for source](#)



A 3x3 grid with a blue path starting at the top-left cell (0,0) and ending at the bottom-right cell (2,2). The path consists of the following cells: (0,0), (0,1), (1,1), (1,2), and (2,2). The cells (0,2), (1,0), and (2,0) are marked with a grey 'X'. The cells (0,1), (1,0), and (1,2) are marked with a grey circle. The cells (1,1) and (2,0) are empty.



A 3D surface plot of the function $f(x_1, x_2)$. The surface is blue and has three distinct peaks. Each peak is marked with an orange dot. The vertical axis is labeled $f(x_1, x_2)$ and ranges from 0 to 0.06. The horizontal axes are labeled x_1 and x_2 , both ranging from -4 to 4.

$$a_1 = 0.2, a_2 = 0.45, a_3 = 0.35$$

$$c_1 = (2, -2), c_2 = (0, 0),$$

$$c_3 = (-3, 2)$$